

Lasso Regression: From First Principles

Why Do We Need Lasso Regression?

Ridge Regression controls overfitting by shrinking coefficients, but it usually keeps all features in the model. Sometimes we want something stronger. We want the model to decide that some features are not useful and reduce their coefficients exactly to zero.

Imagine packing for a trip. Ridge says, "Take everything, but keep the bag light." Lasso says, "Take only what you really need." This makes Lasso useful when we suspect that many features are weak, noisy, or unnecessary.

Lasso Regression is Linear Regression with a penalty that encourages simpler models and can perform automatic feature selection.

The Core Idea

Lasso uses the same prediction equation as Linear Regression.

$$\hat{y} = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \dots + \theta_n x_n$$

The difference is the cost function. Lasso adds a penalty based on the absolute values of coefficients.

$$J(\theta) = MSE + \lambda \sum_{j=1}^n |\theta_j|$$

Here, *MSE* measures prediction error. The term $\sum_{j=1}^n |\theta_j|$ is the L1 penalty. The value λ controls how strong the penalty is.

Understanding the L1 Penalty

The L1 penalty adds the absolute values of coefficients.

$$L1 \text{ Penalty} = |\theta_1| + |\theta_2| + \dots + |\theta_n|$$

Because of the shape of the L1 penalty, Lasso can push some coefficients exactly to zero. When a coefficient becomes zero, that feature no longer affects the prediction.

This is why Lasso can be used for feature selection. It does not just make the model smoother. It can actually remove weak features from the model.

Lambda

The value λ controls how strongly Lasso punishes coefficient size.

If $\lambda = 0$, Lasso becomes ordinary Linear Regression.

If λ is small, only a little shrinkage happens.

If λ is large, many coefficients may become zero.

If λ is too large, the model may remove too much information and underfit.

Choosing λ is about finding the right balance between accuracy and simplicity.

Feature Selection

Feature selection means choosing which features are useful for prediction.

Suppose a model predicts house price using size, location, bedrooms, bathrooms, roof color, and owner birthday. Some features are clearly useful. Others may add noise.

Lasso may learn coefficients like this:

Feature	Coefficient
Size	210
Location Score	50000
Bedrooms	12000
Roof Color	0
Owner Birthday	0

The zero coefficients mean the model has ignored those features. This makes the model simpler and easier to interpret.

Ridge vs Lasso

Ridge uses an L2 penalty.

$$\lambda \sum_{j=1}^n \theta_j^2$$

Lasso uses an L1 penalty.

$$\lambda \sum_{j=1}^n |\theta_j|$$

Ridge shrinks coefficients smoothly and usually keeps all features. Lasso can shrink some coefficients exactly to zero and remove features.

Use Ridge when many features are useful and correlated. Use Lasso when you believe only some features are important and you want automatic feature selection.

Elastic Net

Elastic Net combines Ridge and Lasso. It uses both L1 and L2 penalties.

$$J(\theta) = MSE + \lambda_1 \sum_{j=1}^n |\theta_j| + \lambda_2 \sum_{j=1}^n \theta_j^2$$

This can be useful when there are many correlated features. Lasso may choose one feature from a correlated group and ignore the others. Elastic Net can behave more smoothly by combining feature selection with Ridge-style stability.

Worked Example

Suppose two models have similar error.

Model	MSE	Coefficients
A	12	5, 0, 0, 2
B	10	5, 3, 4, 2

Model B has lower error but uses more features. Lasso may prefer Model A if the extra improvement from Model B is not worth the extra complexity.

This is the Lasso mindset: use the features that matter, ignore the ones that do not add enough value.

Advantages, Disadvantages, and Evaluation

Lasso is useful because it reduces overfitting and performs feature selection. It can make a model easier to interpret by removing unnecessary features.

Its disadvantage is that it can be unstable when features are strongly correlated. If two features contain similar information, Lasso may choose one and remove the other, even though both are meaningful.

Feature scaling is important because Lasso penalizes coefficients. Without scaling, the penalty may treat features unfairly.

Lasso is evaluated using regression metrics such as MAE, MSE, RMSE, and R^2 .

Use Lasso when you want a linear model that can also select features. Avoid relying on Lasso alone when the relationship is highly non-linear or when correlated features make interpretation tricky.